



CHRIST
(DEEMED TO BE UNIVERSITY)
DELHI - NCR, INDIA

Computer Vision

MCA-574

Lab – 03

BY

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SUBMITTED TO

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Remote Sensing: Sampling and Quantization Effects

This notebook demonstrates how spatial sampling and intensity quantization affect image representation, quality, and transmission efficiency for high-resolution satellite images.

Task 1: Load High-Resolution Color Image

We'll use a sample satellite image.

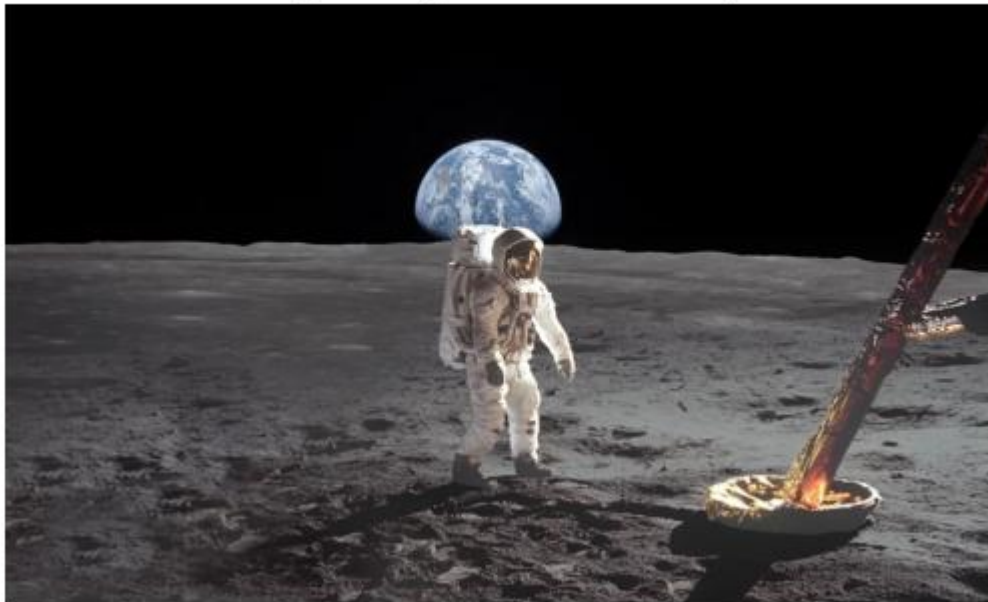
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load high-resolution color image
img = cv2.imread('moon.webp')
if img is None:
    raise Exception('Image not found!')

img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
plt.imshow(img_rgb)
plt.title('Original High-Resolution Image')
plt.axis('off')
plt.show()
```

Output:

Original High-Resolution Image



Task 2: Spatial Sampling (Resolution Reduction)

We will reduce the image resolution stepwise: $512 \times 512 \rightarrow 256 \times 256 \rightarrow 128 \times 128$.

```
# Resize image to different resolutions
img_512 = cv2.resize(img_rgb, (512, 512))
img_256 = cv2.resize(img_rgb, (256, 256))
img_128 = cv2.resize(img_rgb, (128, 128))

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img_512)
plt.title('512x512')
plt.axis('off')

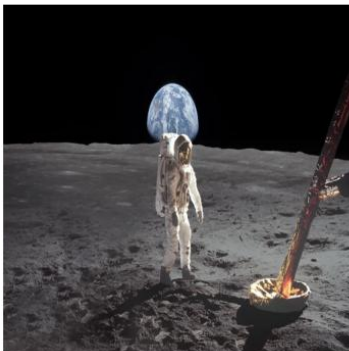
plt.subplot(1,3,2)
plt.imshow(img_256)
plt.title('256x256')
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(img_128)
plt.title('128x128')
plt.axis('off')

plt.show()
```

Output:

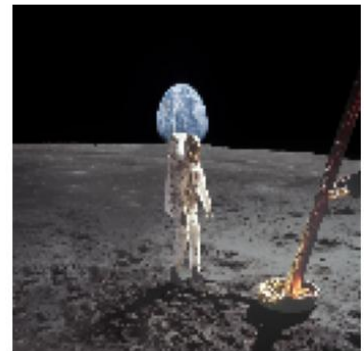
512x512



256x256



128x128



Task 3: Intensity Quantization (Gray Level Reduction)

We will reduce the number of gray levels: $256 \rightarrow 64 \rightarrow 16$.

```
# Convert to grayscale for quantization
gray_512 = cv2.cvtColor(img_512, cv2.COLOR_RGB2GRAY)

# Quantize to 256, 64, and 16 levels
def quantize(img, levels):
    step = 256 // levels
    return (img // step) * step

q_256 = quantize(gray_512, 256)
q_64 = quantize(gray_512, 64)
q_16 = quantize(gray_512, 16)

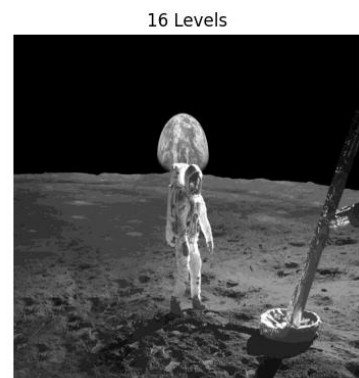
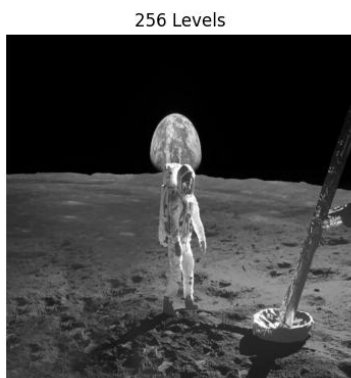
plt.figure(figsize=(15,5))
plt.subplot(1,3,1)
plt.imshow(q_256, cmap='gray')
plt.title('256 Levels')
plt.axis('off')

plt.subplot(1,3,2)
plt.imshow(q_64, cmap='gray')
plt.title('64 Levels')
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(q_16, cmap='gray')
plt.title('16 Levels')
plt.axis('off')

plt.show()
```

Output:



Task 4: Visual Quality & Data Efficiency Comparison

Let's compare the images visually and discuss data storage/transmission efficiency.

```
# Calculate and print image sizes (in bytes)
import sys
sizes = [img_512.nbytes, img_256.nbytes, img_128.nbytes]
quant_sizes = [q_256.nbytes, q_64.nbytes, q_16.nbytes]

print('Resolution Data Sizes (bytes):')
print(f'512x512: {sizes[0]} | 256x256: {sizes[1]} | 128x128: {sizes[2]}')
print('Quantized Data Sizes (bytes):')
print(f'256 levels: {quant_sizes[0]} | 64 levels: {quant_sizes[1]} | 16 levels: {quant_sizes[2]}')
```

Output :

Resolution Data Sizes (bytes):

512x512: 786432 | 256x256: 196608 | 128x128: 49152

Quantized Data Sizes (bytes):

256 levels: 262144 | 64 levels: 262144 | 16 levels: 262144

Task 5: Questions & Discussion

Q1: Trade-off between image quality and data size

- Reducing resolution and quantization levels decreases data size, but also degrades visual quality. Essential details may be lost if levels are too low.

Q2: Importance of selecting appropriate sampling and quantization levels

- For applications needing fine detail (e.g., urban mapping, crop analysis), higher resolution and more gray levels are needed.
- For quick transmission or rough monitoring (e.g., weather, IoT sensors), lower levels may suffice.

Q3: Real-world implications for satellite imaging or IoT camera-based systems

- Satellite images for scientific analysis require high fidelity, but bandwidth is limited. Adaptive compression (region-based, multi-resolution) is often used.
- IoT cameras may use aggressive quantization/sampling to save power and bandwidth, but must balance with the need for actionable information.

Summary Table:

Setting	Data Size	Visual Quality	Use Case Example
High Resolution, 256 Levels	Large	Excellent	Scientific mapping
Low Resolution, 16 Levels	Small	Poor	Quick event detection
Medium, 64 Levels	Moderate	Good	General monitoring